Luxury Housing Sales Analysis – Bengaluru

Project Overview

This project focuses on analyzing luxury housing data in Bengaluru to uncover key business insights related to builder performance, pricing trends, booking behavior, and market demand.

The solution integrates:

- **Data Cleaning using Python (Pandas)**
- **Data Storage & Querying using SQL**
- **Interactive Dashboard using Power**

Objectives

- Identify top-performing builders based on revenue and booking success
- Analyze booking conversion trends across micro-markets
- Understand the impact of amenities on booking decisions
- Evaluate effectiveness of sales channels
- Identify most in-demand housing configurations
- Visualize geographical distribution of projects

Data Cleaning & Preprocessing (Python)

Performed using **Pandas & NumPy**:

- Removed duplicates
- Handled missing values
- Converted data types
- Standardized column names
- Cleaned numerical fields (ticket_price, amenity_score)
- Filled missing categorical values

SQL Analysis

Key queries used:

✓ **Builder Performance**

- Total revenue
- Average ticket size

✓ **Amenity Impact**

- Relationship between amenity score & booking conversion

✓ **Booking Conversion**

- Booked vs not booked by micro-market

✓ **Configuration Demand**

- Most popular housing types

✓ **Sales Channel Efficiency**

- Performance of different sales channels

✓ **Quarterly Builder Contribution**

- Revenue trend across quarters

✓ **Geographical Insights**

- Project concentration by micro-market

Power BI Dashboard

The dashboard includes:

🔹 **KPI Section**

- Total Revenue
- Successful Bookings
- Booking Success Rate

- **Builder Analysis**
  - Top builders by revenue (Bar Chart)
  - Quarterly revenue matrix

- **Customer & Market Insights**
  - Amenity vs Booking Conversion (Scatter Plot)
  - Booking Conversion by Micro-Market (Stacked Chart)
  - Sales Channel Performance (100% Stacked Chart)

- **Demand Analysis**
  - Configuration Demand (Donut Chart)

- **Geographical Analysis**
  - Map showing project distribution

Conclusion

This project demonstrates an end-to-end data analytics workflow, from raw data cleaning to business insight generation through interactive dashboards. It provides valuable insights for real estate developers and investors to optimize pricing, marketing, and product strategies.